

2.2.1 Connectives

Connectives are the tools for building new formulas from existing ones. We briefly mention them:

Logical negation $\neg p$ ('not p ') is the negation of the formula p

Logical conjunction $p \wedge q$ (' p and q ')

Logical disjunction $p \vee q$ (' p or q ')

Logical implication $p \Rightarrow q$ (' p implies q ' or 'if p , then q ')

Logical equivalence $p \Leftrightarrow q$ (' p is logically equivalent to q ')

Proof by contradiction

This formalism allows us to understand the notion of a proof by contradiction, which is summed up by the logical equivalence:

$$(p \Rightarrow q) \quad \Leftrightarrow \quad (p \wedge \neg q \Rightarrow \neg p)$$

2.2.2 Predicates

A **predicate** is a formula that depends on one or more variables. In fact, we have seen predicates before, when we called them 'characteristic properties'. Here are some more examples:

$p(x)$ = ' x is a prime number'

$q(y)$ = ' y is the square of a natural number'

$r(x, y)$ = ' x is divisible by y '

2.2.3 Quantifiers

In a set X , for a given predicate $p(x)$ with $x \in X$, we can ask whether p is always true, or perhaps only sometimes. This is expressed mathematically as follows:

Universal quantifier: $\forall x, p(x)$ (we say 'for every x , $p(x)$ holds')

Existential quantifier: $\exists x, p(x)$ (we say 'there exists x , such that $p(x)$ holds')

Unique existential quantifier: $\exists! x, p(x)$ (we say 'there exists one and *only one* x , such that $p(x)$ holds')

Example 2.1: Suppose that, as above, $p(x)$ = ' x is a prime number'. If $X = \mathbb{N}$, then it is true that *there exists* $x \in X$ that is a prime number, i.e. $\exists x, p(x)$. However, it is *not true* that *every* $x \in X$ is a prime number. That is, $\neg(\forall x, p(x))$.

Example 2.2: Consider the predicate $p(x) = 'x^2 = x'$. If $X = \{1, 2, 3, \dots\}$, then $1 \in X$ is the unique element in X for which $p(x)$ is true. That is, $\exists! x, p(x)$. On the set $Y = \{2, 3, 4, \dots\}$ the predicate $p(x)$ is never true, i.e. $\neg(\exists x, p(x))$.

2.3 Sets of numbers

2.3.1 The set $\mathbb{N}$ of natural numbers

The set $\mathbb{N}$ of natural numbers satisfies an important axiom:

Well-Ordering Principle

The set $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ satisfies the following *axiom*:

The Well-Ordering Principle. Every non-empty subset $A \subseteq \mathbb{N}$ contains a least ('first') element.

This allows us to define the idea of *induction*:

Theorem 2.3 (Principle of Induction): Let $N \in \mathbb{N}$ and denote by $p(n)$ a predicate defined for every $n \geq N$, $n \in \mathbb{N}$. Suppose that the following hold:

1. $p(N)$ is true,
2. $\forall n \geq N, p(n) \Rightarrow p(n+1)$.

Then $p(n)$ is true for all integers $n \geq N$.

Proof. By contradiction, assume that $\exists n \geq N$ for which $p(n)$ is false. Then the set

$$F = \{n \in \mathbb{N} \mid n \geq N \text{ and } p(n) \text{ is false}\}$$

is not empty. By the well-ordering principle of $\mathbb{N}$, define $m \in F$ to be the smallest number in F . Then $p(m)$ is false. Therefore $m \neq N$ (recall that we know that $p(N)$ is true). So necessarily $m > N$, and it follows that $m-1 \geq N$. By our definition of the number m , $p(m-1)$ must be true (otherwise $m-1$ would have been the smallest number in F). But we know that $\forall n \geq N, p(n) \Rightarrow p(n+1)$. Taking $n = m-1$ we get that $p(m-1) \Rightarrow p(m)$. But this is not true, since $p(m-1)$ is true while $p(m)$ is false. We have therefore reached a contradiction, so that $\neg(\exists n \geq N \text{ for which } p(n) \text{ is false})$, i.e. $\forall n \geq N, p(n)$ is true. $\square$

Proof by induction

To prove that a statement $p(n)$ is true for all $n \geq N$ by induction, we need to demonstrate two things:

1. that $p(N)$ (the **base case**) is true,
2. that $p(n)$ being true (the **induction assumption**) implies $p(n+1)$ being true for all $n \geq N$.

Example 2.3 (Bernoulli inequality): We claim that $\forall r \geq -1$, the *Bernoulli inequality*

$$(1+r)^n \geq 1+nr, \quad \forall n \in \mathbb{N},$$

holds. We prove this by induction. Here

$$p(n) = '(1 + r)^n \geq 1 + nr'.$$

1. For $n = 0$, we have $(1 + r)^0 = 1$ and $1 + 0 \cdot r = 1$ so that $(1 + r)^0 \geq 1 + 0 \cdot r$ and therefore $p(0)$ is true.

2. Now assume that $p(n)$ is true. This is called the **induction assumption**. Let us show that $p(n + 1)$ is true. Using the fact that $1 + r \geq 0$, we have

$$\begin{aligned} (1 + r)^{n+1} &= (1 + r)(1 + r)^n \\ &\geq (1 + r)(1 + nr) \quad (\text{here we use the induction assumption and that } 1 + r \geq 0) \\ &= 1 + (n + 1)r + nr^2 \\ &\geq 1 + (n + 1)r. \quad (\text{since } nr^2 \geq 0) \end{aligned}$$

Hence $p(n + 1)$ is true, and by the Principle of Induction (we usually just say ‘*by induction*’) the Bernoulli inequality holds for all $n \in \mathbb{N}$.

2.3.2 The set $\mathbb{Q}$ of rational numbers

To simplify notation, we first define the set

$$\mathbb{N}_+ = \mathbb{N} \setminus \{0\} = \{1, 2, 3, 4, \dots\} = \text{set of positive integers.}$$

We now have the tools to define the set $\mathbb{Q}$ as:

$$\mathbb{Q} = \left\{ r = \frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{N}_+ \right\}$$

Any rational number r has infinitely many representations. For example

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \dots$$

Therefore, we normally choose the (unique) representative such that p and q have no common divisors.

We can also write fractions in base 10 (which is the standard base for us), as

$$r = \pm(c_k 10^k + c_{k-1} 10^{k-1} + \dots + c_1 10 + c_0 + c_{-1} 10^{-1} + c_{-2} 10^{-2} + \dots)$$

where all coefficients $c_k, c_{k-1}, \dots, c_0, c_{-1}, \dots$ can assume any of the values $0, 1, 2, \dots, 9$. Here are some examples:

$$\begin{aligned} \frac{1}{2} &= +(0 \cdot 10^k + \dots + 0 \cdot 10 + 0 + 5 \cdot 10^{-1} + 0 \cdot 10^{-2} + \dots) = 0.5, \\ -\frac{29}{2} &= -14\frac{1}{2} = -(0 \cdot 10^k + \dots + 1 \cdot 10 + 4 + 5 \cdot 10^{-1} + 0 \cdot 10^{-2} + \dots) = -14.5, \\ \frac{1}{3} &= +(0 \cdot 10^k + \dots + 0 \cdot 10 + 0 + 3 \cdot 10^{-1} + 3 \cdot 10^{-2} + \dots) = 0.\overline{33}, \end{aligned}$$

where the overline means that the expression repeats itself infinitely:

$$\overline{a_1 a_2 \dots a_m} = (a_1 a_2 \dots a_m)(a_1 a_2 \dots a_m)(a_1 a_2 \dots a_m) \dots$$

Decimal representation of rational numbers

Using long division it can be verified that the decimal representation of any rational number is either finite (as in the example for $\frac{1}{2}$) or infinitely repeating (as in the example for $\frac{1}{3}$). The converse is also true: every number that has a finite or repeating representation is rational.

2.3.3 The set $\mathbb{R}$ of real numbers

The world around us is made up of three space dimensions and one time dimension which are all continuous. We move through space and time in a continuous fashion. So, a natural question is whether the rational numbers are enough to describe our world? For instance, as time elapses from 1pm to 2pm, can all intermediate moments be described by rational numbers?

The answer turns out to be *no*. Between 1 and 2, for instance, there exists a number which is the solution of the equation $x^2 = 2$ (we call this number *the square root of 2* and denote it by $\sqrt{2}$) which is not rational; we normally say that it is **irrational**.

Lemma 2.4 ($\sqrt{2}$ is irrational): The number x satisfying the equation $x^2 = 2$ is irrational.

Proof. We prove this by contradiction. Suppose that there is a rational solution $x = \frac{p}{q}$ (with p and q having no common divisors). Then $\frac{p^2}{q^2} = 2$ so that

$$p^2 = 2q^2.$$

Hence p^2 is even (it is divisible by 2). If p^2 is even then so is p itself. Hence $\exists k \in \mathbb{N}$ such that $p = 2k$ (observe that, in fact, k must be positive, otherwise $x = 0$). We therefore have $(2k)^2 = 2q^2$ which implies that

$$2k^2 = q^2.$$

It follows that q^2 is even, and consequently so is q itself. But this means that p and q have the number 2 as a common divisor, a contradiction. Therefore there is no rational representation for x . $\square$

The real number line. We can write all rational numbers along a single line, placing bigger numbers to the right. Since $1^2 = 1$ and $2^2 = 4$, it seems obvious that $\sqrt{2}$ must lie between 1 and 2. We have thus found an irrational number between 1 and 2,

$$1 < \sqrt{2} < 2.$$

Since it is irrational, it has an infinite representation that *never* repeats. Therefore it is *impossible* to exactly write its value. The beginning of its expansion is:

$$\sqrt{2} = 1.41421356 \dots$$

It follows that

$$\begin{aligned} 1.4 &< \sqrt{2} < 1.5 \\ 1.41 &< \sqrt{2} < 1.42 \\ 1.414 &< \sqrt{2} < 1.415 \\ 1.4142 &< \sqrt{2} < 1.4143 \\ &\vdots \end{aligned}$$

These are sequences of rational numbers to the left and to the right of $\sqrt{2}$, both of which tend to $\sqrt{2}$, but never reach it. So $\sqrt{2}$ fills a certain gap. Indeed, with the irrational numbers added, we get a continuous line of increasing numbers, called the *real number line*. This property of having a *continuous* line is called *completeness* and we say that the real number line is **complete** (intuitively, it means that there are no gaps). This is in fact an axiom (called The Completeness Axiom), discussed later in the chapter.

Actually, there are *many* irrationals. It turns out that there are infinitely many rationals and infinitely many irrationals. They are all intertwined:

- between any two rationals $r_1 < r_2$ there are infinitely many irrationals,
- between any two irrationals $y_1 < y_2$ there are infinitely many rationals.

However, there are *more* irrationals than there are rationals. If we write the set of real numbers (the real number line) as the (disjoint) union of the rationals and the irrationals

$$\mathbb{R} = \mathbb{Q} \sqcup (\mathbb{R} \setminus \mathbb{Q})$$

then both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are infinite, but $\mathbb{R} \setminus \mathbb{Q}$ is ‘bigger’ in a certain sense. This statement is made rigorous using a mathematical theory called *measure theory*. It can be understood intuitively as follows: if a number is chosen randomly between, say, 0 and 1, then it is *almost surely* an irrational number (in other words, the probability of choosing a rational number is zero). So within the real number line there are ‘more’ irrationals. However there are also rationals *everywhere*. We say that **the rational numbers $\mathbb{Q}$ are dense within the reals $\mathbb{R}$.**

The ordering of real numbers

The real numbers are a **totally ordered set**: $\forall x, y \in \mathbb{R}$, one (and *only* one) of the following properties holds:

$$x = y \quad \text{or} \quad x < y \quad \text{or} \quad x > y.$$

We often use the following symbols for these important subsets of $\mathbb{R}$:

$$\begin{aligned} \mathbb{R}_+ &= \{x \in \mathbb{R} \mid x > 0\} \\ \mathbb{R}_- &= \{x \in \mathbb{R} \mid x < 0\} \end{aligned}$$